

MANSUN CHAN

professor

基本信息

姓名: MANSUN CHAN

所属院校: 香港科技大学

邮箱: mchan@ust.hk

地区: China-Hong Kong

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可, 展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

集成电路设计与集成系统

材料物理

电子科学与技术

研究领域

微电子器件与制造

纳米电子与先进工艺

低维材料与量子效应

半导体与光电子材料

微纳电子与纳米技术

半导体器件物理与工艺

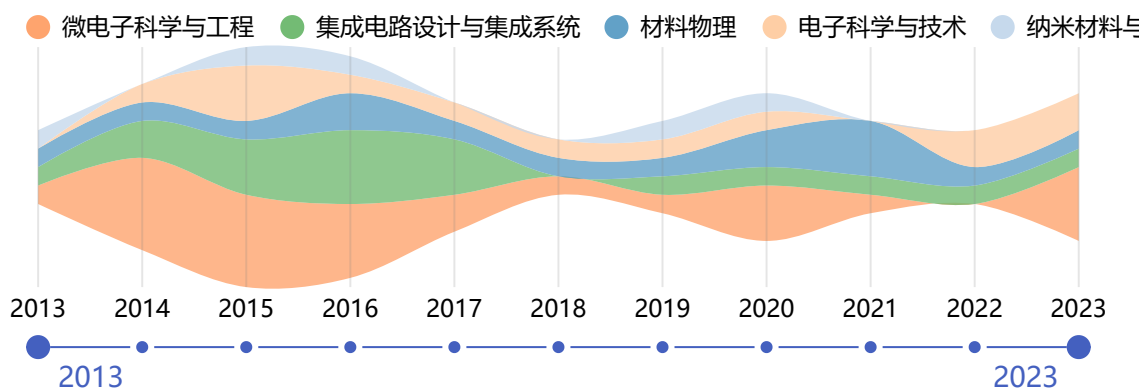
低功耗电路与能量管理

个人简介

Prof. Mansun Chan, Alex Wong Siu Wah Gigi Wong Fook Chi Professor of Engineering at the Hong Kong University of Science and Technology, holds a PhD in Electrical Engineering from the University of California, Berkeley. A Fellow of IEEE and IEEE Distinguished Lecturer, his research focuses on Integrated Circuits and Systems, Microelectronics, and Quantum Engineering, with particular interests in Nano-CMOS Technology, 2-D devices and circuits, AI technologies and systems, and Quantum computing. Prof. Chan is also the

Director of the Academy of Continuing Education and the MSc program in IC Design Engineering at HKUST.

近年发表的论文



Generic Compact Modeling of Emerging Memories With Recurrent NARX Network

IEEE ELECTRON DEVICE LETTERS

Rong, Zhao; Dai, Wu; Feng, Ning; Yang, Yuhang; Zhang, Lining; Wang, Zongwei; Cai, Yimao; Chan, Mansun

被引用次数: 0

2023

Deep Sub-Micron Self-Aligned Bottom-Gate Amorphous InGaZnO Thin-Film Transistors With Low-Resistance Source/Drain

IEEE ELECTRON DEVICE LETTERS

Zhang, Yuhang; Li, Jiye; Zhang, Yuqing; Yang, Huan; Guan, Yuhang; Chan, Mansun; Lu, Lei; Zhang, Shengdong

被引用次数: 0

2023

Ultra-Flexible Monolithic 3D Complementary Metal-Oxide-Semiconductor Electronics

ADVANCED FUNCTIONAL MATERIALS

Zhang, Jiaona; Wang, Wantings; Zhu, Jiahao; Wang, Jialiang; Zhao, Changbin; Zhu, Tianyu; Ren, Qinqi; Liu, Qi; Qiu, Rui; Zhang, Min; Wang, Xinwei; Meng, Hong; Chang, Kuan-Chang; Zhang, Shengdong; Chan, Mansun

被引用次数: 4

2023

Threshold Voltage Model for 2-D FETs With Undoped Body and Gated Source

IEEE TRANSACTIONS ON ELECTRON DEVICES

Estrada, Cristine Jin; Ma, Zichao; Zhang, Lining; Chan, Mansun

被引用次数: 1

2023

3-Masks-Processed Sub-100 nm Amorphous InGaZnO Thin-Film Transistors for Monolithic 3D Capacitor-Less Dynamic Random Access Memories

ADVANCED ELECTRONIC MATERIALS

Zhang, Yuqing; Li, Jiye; Li, Jinxiong; Huang, Tengyan; Guan, Yuhang; Zhang, Yuhang; Yang, Huan; Chan, Mansun; Wang, Xinwei; Lu, Lei; Zhang, Shengdong

被引用次数: 0

2023

On-Chip Integrated High Gain Complementary MoS₂ Inverter Circuit with Exceptional High Hole Current P-Channel Field-Effect Transistors

ADVANCED ELECTRONIC MATERIALS

Ma, Zichao; Estrada, Cristine Jin; Gong, Kui; Zhang, Lining; Chan, Mansun

被引用次数: 6

2022

A neuromorphic core based on threshold switching memristor with asynchronous address event representation circuits

SCIENCE CHINA-INFORMATION SCIENCES

Wei, Jinsong; Zhang, Jilin; Zhang, Xumeng; Wu, Zuheng; Wang, Rui; Lu, Jian; Shi, Tuo; Chan, Mansun; Liu, Qi; Chen, Hong

被引用次数: 3

2022

Transferred metal gate to 2D semiconductors for sub-1 V operation and near ideal subthreshold slope

SCIENCE ADVANCES

Wang, Jingli; Cai, Lejuan; Chen, Jiewei; Guo, Xuyun; Liu, Yuting; Ma, Zichao; Xie, Zhengdao; Huang, Hao; Chan, Mansun; Zhu, Ye; Liao, Lei; Shao, Qiming; Chai, Yang

被引用次数: 40

2021

High Current Nb-Doped P-Channel MoS₂ Field-Effect Transistor Using Pt Contact

IEEE ELECTRON DEVICE LETTERS

Ma, Zichao; Zhang, Lining; Zhou, Changjian; Chan, Mansun

被引用次数: 8

2021

Complementary Two-Dimensional (2-D) FET Technology With MoS₂/hBN/Graphene Stack

IEEE ELECTRON DEVICE LETTERS

Estrada, Cristine Jin; Ma, Zichao; Chan, Mansun

被引用次数: 6

2021

Air-Gap Technology With a Large Void-Fraction for Global Interconnect Delay Reduction

IEEE TRANSACTIONS ON ELECTRON DEVICES

Prawoto, Clarissa; Ma, Zichao; Xiao, Ying; Raju, Salahuddin; Chan, Mansun

被引用次数: 3

2021

Hybrid memristor-CMOS neurons for in-situ learning in fully hardware memristive spiking neural networks

SCIENCE BULLETIN

Zhang, Xumeng; Lu, Jian; Wang, Zhongrui; Wang, Rui; Wei, Jinsong; Shi, Tuo; Dou, Chunmeng; Wu, Zuheng; Zhu, Jiaxue; Shang, Dashan; Xing, Guozhong; Chan, Mansun; Liu, Qi; Liu, Ming

被引用次数: 60

2021

A SPICE Model of Phase Change Memory for Neuromorphic Circuits

IEEE ACCESS

Chen, Xuhui; Hu, Huifang; Huang, Xiaoqing; Cai, Weiran; Liu, Ming; Lam, Chung; Lin, Xinnan; Zhang, Lining; Chan, Mansun

被引用次数: 12

2020

A Compact Phase Change Memory Model With Dynamic State Variables

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hu, Huifang; Liu, Dayong; Chen, Xuhui; Dong, Deqi; Cui, Xiaole; Liu, Ming; Lin, Xinnan; Zhang, Lining; Chan, Mansun

被引用次数: 11

2020

Synthesis of Vertical Carbon Nanotube Interconnect Structures Using CMOS-Compatible Catalysts

NANOMATERIALS

Ma, Zichao; Zhou, Shaolin; Zhou, Changjian; Xiao, Ying; Li, Suwen; Chan, Mansun

被引用次数: 8

2020

Self-driven WSe₂ photodetectors enabled with asymmetrical van der Waals contact interfaces

NPJ 2D MATERIALS AND APPLICATIONS

Zhou, Changjian; Zhang, Shouyong; Lv, Zhe; Ma, Zichao; Yu, Cui; Feng, Zhihong; Chan, Mansun

被引用次数: 60

2020

A Compact Model for Stochastic Spike-Timing-Dependent Plasticity (STDP) Based on Resistive Switching Memory (RRAM) Synapses

IEEE TRANSACTIONS ON ELECTRON DEVICES

Estrada, Cristine Jin; Xiao, Ying; Xu, Chen; Chan, Mansun

被引用次数: 6

2020

Analytical Monolayer MoS₂ MOSFET Modeling Verified by First Principle Simulations

IEEE ELECTRON DEVICE LETTERS

Ahmed, Zubair; Shi, Qing; Ma, Zichao; Zhang, Lining; Guo, Hong; Chan, Mansun

被引用次数: 10

2020

Interconnect Technology With h-BN-Capped Air-Gaps

IEEE ELECTRON DEVICE LETTERS

Prawoto, Clarissa; Ma, Zichao; Xiao, Ying; Raju, Salahuddin; Zhou, Changjian; Chan, Mansun

被引用次数: 5

2019

Low Temperature Synthesis of High-Density Carbon Nanotubes on Insulating Substrate

NANOMATERIALS

Xiao, Ying; Ahmed, Zubair; Ma, Zichao; Zhou, Changjian; Zhang, Lining; Chan, Mansun

被引用次数: 6

2019

A Dynamic Time Evolution Method for Concurrent Device-Circuit Aging Simulations

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Lining; Ma, Chenyue; Xiao, Ying; Zhang, Hanyu; Lin, Xinnan; Chan, Mansun

被引用次数: 4

2019

On-Demand Band-Rejected Wideband Antenna Based on Peelable Resonator Membrane

IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS

Wang, Rong; Li, Min; Raju, Salahuddin; Roberts, Robert C.; Chan, Mansun; Jiang, Lijun

被引用次数: 1

2018

Self-Driven Metal-Semiconductor-Metal WSe₂ Photodetector with Asymmetric Contact Geometries

ADVANCED FUNCTIONAL MATERIALS

Zhou, Changjian; Raju, Salahuddin; Li, Bin; Chan, Mansun; Chai, Yang; Yang, Cary Y.

被引用次数: 137

2018

Origin of Nonideal Graphene-Silicon Schottky Junction

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Xintong; Zhang, Lining; Ahmed, Zubair; Chan, Mansun

被引用次数: 10

2018

A 1.2V-to-0.4V 3.2GHz-to-14.3MHz Power-Efficient 3-Port Register File in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Sarfraz, Khawar; Chan, Mansun

被引用次数: 4

2017

A Low-Power Ring Oscillator Using Pull-Up Control Scheme Integrated by Metal-Oxide TFTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wu, Jian-Dong; Zhan, Fan; Zhou, Lei; Wu, Wei-Jing; Xu, Miao; Wang, Lei; Yao, Ruo-He; Peng, Jun-Biao; Chan, Mansun

被引用次数: 20

2017

Doping of two-dimensional MoS₂ by high energy ion implantation

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Xu, Kang; Zhao, Yuda; Lin, Ziyuan; Long, Yan; Wang, Yi; Chan, Mansun; Chai, Yang

被引用次数: 35

2017

Tunneling contact IGZO TFTs with reduced saturation voltages

APPLIED PHYSICS LETTERS

Wang, Longyan; Sun, Yin; Zhang, Xintong; Zhang, Lining; Zhang, Shengdong; Chan, Mansun

被引用次数: 27

2017

A 140-mV Variation-Tolerant Deep Sub-Threshold SRAM in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sarfraz, Khawar; He, Jin; Chan, Mansun

被引用次数: 12

2017

Analytical Current Model for Long-Channel Junctionless Double-Gate MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Xinnan; Zhang, Baili; Xiao, Ying; Lou, Haijun; Zhang, Lining; Chan, Mansun

被引用次数: 17

2016

Modeling CNTFET Performance Variation Due to Spatial Distribution of Carbon Nanotubes

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ahmed, Zubair; Zhang, Lining; Sarfraz, Khawar; Chan, Mansun

被引用次数: 3

2016

Carbon Nanotube Contact Plug on Silicide for CMOS Compatible Interconnect

IEEE ELECTRON DEVICE LETTERS

Li, Suwen; Raju, Salahuddin; Zhou, Changjian; Chan, Mansun

被引用次数: 11

2016

Doping enhanced barrier lowering in graphene-silicon junctions

APPLIED PHYSICS LETTERS

Zhang, Xintong; Zhang, Lining; Chan, Mansun

被引用次数: 10

2016

Carrier Type Control of WSe₂ Field-Effect Transistors by Thickness Modulation and MoO₃ Layer Doping

ADVANCED FUNCTIONAL MATERIALS

Zhou, Changjian; Zhao, Yuda; Raju, Salahuddin; Wang, Yi; Lin, Ziyuan; Chan, Mansun; Chai, Yang

被引用次数: 190

2016

A Short Channel Double-Gate Junctionless Transistor Model Including the Dynamic Channel Boundary Effect

IEEE TRANSACTIONS ON ELECTRON DEVICES

Xiao, Ying; Lin, Xinnan; Lou, Haijun; Zhang, Baili; Zhang, Lining; Chan, Mansun

被引用次数: 30

2016

A Compact Model for Double-Gate Heterojunction Tunnel FETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Dong, Yunpeng; Zhang, Lining; Li, Xiangbin; Lin, Xinnan; Chan, Mansun

被引用次数: 36

2016

Comparative Analysis of Carrier Statistics on MOSFET and Tunneling FET Characteristics

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Li, Dan; Zhang, Baili; Lou, Haijun; Zhang, Lining; Lin, Xinnan; Chan, Mansun

被引用次数: 4

2015

Eight-FinFET Fully Differential SRAM Cell With Enhanced Read and Write Voltage Margins

IEEE TRANSACTIONS ON ELECTRON DEVICES

Salahuddin, Shairfe Muhammad; Chan, Mansun

被引用次数: 23

2015

Gate Capacitance Model for Aligned Carbon Nanotube FETs With Arbitrary CNT Spacing

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ahmed, Zubair; Zhang, Lining; Chan, Mansun

被引用次数: 6

2015

Suppression of subthreshold characteristics variation for junctionless multigate transistors using high-k spacers

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Lou, Haijun; Zhang, Baili; Li, Dan; Lin, Xinnan; He, Jin; Chan, Mansun

被引用次数: 9

2015

Resistance Determination for Sub-100-nm Carbon Nanotube Vias

IEEE ELECTRON DEVICE LETTERS

Zhou, Changjian; Vyas, Anshul A.; Wilhite, Patrick; Wang, Phillip; Chan, Mansun; Yang, Cary Y.

被引用次数: 22

2015

Low voltage and high ON/OFF ratio field-effect transistors based on CVD MoS₂ and ultra high-k gate dielectric PZT

NANOSCALE

Zhou, Changjian; Wang, Xinsheng; Raju, Salahuddin; Lin, Ziyuan; Villaroman, Daniel; Huang, Baoling; Chan, Helen Lai-Wa; Chan, Mansun; Chai, Yang

被引用次数: 134

2015

Extraction and modeling of layout-dependent MOSFET gate-to-source/drain fringing capacitance in 40 nm technology

SOLID-STATE ELECTRONICS

Sun, Lijie; Shang, Ganbing; Liu, Linlin; Cheng, Jia; Guo, Ao; Ren, Zheng; Hu, Shaojian; Chen, Shoumian; Zhao, Yuhang; Chan, Mansun; Zhang, Long; Li, Xiaojin; Shi, Yanling

被引用次数: 7

2015

Modeling of Mutual Coupling Between Planar Inductors in Wireless Power Applications

IEEE TRANSACTIONS ON POWER ELECTRONICS

Raju, Salahuddin; Wu, Rongxiang; Chan, Mansun; Yue, C. Patrick

被引用次数: 306

2014

Analytical Modeling of Capacitances for GaN HEMTs, Including Parasitic Components

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Aixi; Zhang, Lining; Tang, Zhikai; Cheng, Xiaoxu; Wang, Yan; Chen, Kevin J.; Chan, Mansun

被引用次数: 40

2014

3-D Resistance Model for Phase-Change Memory Cell

IEEE TRANSACTIONS ON ELECTRON DEVICES

Chen, Yihan; Kwong, Kit Chu; Lin, Xinnan; Song, Zhitang; Chan, Mansun

被引用次数: 3

2014

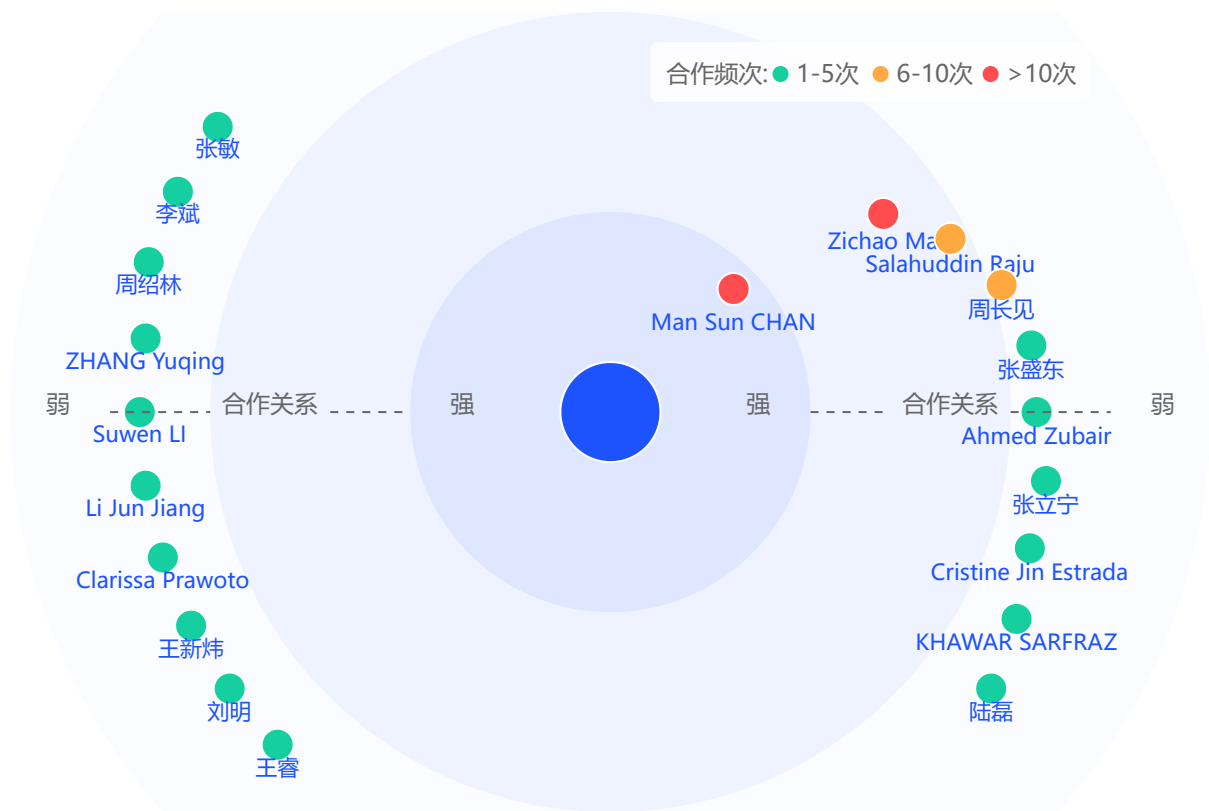
Implementation of Fully Self-Aligned Homo Junction Double-Gate a-IGZO TFTs

IEEE ELECTRON DEVICE LETTERS

He, Xin; Wang, Longyan; Xiao, Xiang; Deng, Wei; Zhang, Letao; Chan, Mansun; Zhang, Shengdong

被引用次数: 23	2014
A Hybrid a-Si and Poly-Si TFTs Technology for AMOLED Pixel Circuits	
JOURNAL OF DISPLAY TECHNOLOGY	
Wang, Longyan; Sun, Lei; Han, Dedong; Wang, Yi; Chan, Mansun; Zhang, Shengdong	
被引用次数: 7	2014
SPICE Modeling of Double-Gate Tunnel-FETs Including Channel Transports	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Zhang, Lining; Chan, Mansun	
被引用次数: 73	2014
Model Order Reduction for Quantum Transport Simulation of Band-To-Band Tunneling Devices	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Huang, Jun Z.; Zhang, Lining; Chew, Weng Cho; Yam, Chi-Yung; Jiang, Li Jun; Chen, Guan-Hua; Chan, Mansun	
被引用次数: 4	2014
Suppression of tunneling leakage current in junctionless nanowire transistors	
SEMICONDUCTOR SCIENCE AND TECHNOLOGY	
Lou, Haijun; Li, Dan; Dong, Yan; Lin, Xinnan; He, Jin; Yang, Shengqi; Chan, Mansun	
被引用次数: 16	2013
First Principles Simulations of Nanoscale Silicon Devices With Uniaxial Strain	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Zhang, Lining; Zahid, Ferdows; Zhu, Yu; Liu, Lei; Wang, Jian; Guo, Hong; Chan, Philip Ching Ho; Chan, Mansun	
被引用次数: 13	2013

学术合作



Man Sun CHAN

香港科技大学

合作次数: 22

Zichao Ma

香港科技大学

合作次数: 11

Salahuddin Raju

香港科技大学

合作次数: 8

周长见

华南理工大学

合作次数: 6

张盛东

北京大学

合作次数: 5

还有 43 个合作者